HIF-P4H-2 inhibition in experimental glaucoma

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Background: Oxygen plays a central role in the pathogenesis of retinal neurodegeneration. The high metabolic rate and O_2 consumption make retina vulnerable to oxidative stress and hypoxia. Hypoxia-inducible factor (HIF) mediates an adaptive response to oxygen deficiency by activating compensatory pathways that both increase O_2 delivery and decrease its consumption. HIF prolyl 4-hydroxylase-2 (HIF-P4H-2/PHD2/EGLN1) is the primary regulator of HIF. HIF-P4H-2 requires O_2 for catalysis but has a low affinity for it, making it an O_2-sensor. We hypothesize that moderate stabilization of HIF by HIF-P4H-2 inhibition protects from retinal ganglion cell (RGC) death and optic nerve damage seen in glaucoma.

Methods: Two different methods of HIF-P4H-2 inhibition were used: A mouse line with genetic repression of *Hif-p4h-2* (*Hif-p4h-2*$^{gt/gt}$) achieved by insertion of a beta-galactosidase reporter into truncated *Hif-p4h-2*, leading to HIF-P4H-2 hypomorphism, and second, systemically administered roxadustat, a pan-HIF-P4H inhibitor. X-Gal staining, qPCR and Western blot were used to determine HIF-P4H-2 and HIF levels at baseline. Experimental glaucoma was induced unilaterally by acute ocular hypertension (AOHT) or by optic nerve crush (ONC). Retinal neurodegeneration was quantified with Brn3a immunohistochemistry in retinal flat mounts and histology was evaluated from HE-stained cross-sections and optic nerves.

Results: *Hif-p4h-2*$^{gt/gt}$ mice had similar RGC survival with wild-type mice in the ONC model but showed significantly higher RGC survival compared to wild-type mice in the AOHT model. *Hif-p4h-2*$^{gt/gt}$ mice also showed preserved tissue integrity compared to wild type in the AOHT model. The mice treated with roxadustat had similar RGC survival compared to vehicle group in the AOHT model, however, induced hemoglobin levels in the roxadustat-treated mice correlated with higher RGC survival.

Conclusions: Genetic HIF-P4H-2 deficiency protects RGCs from ischemia-reperfusion injury in the AOHT model but does not protect RGCs from direct axonal injury in the ONC model. Systemically administered roxadustat does not offer similar level of protection, although induced hypoxia signaling correlates with higher RGC survival, suggesting that HIF stabilization must be local to effectively protect against glaucoma pathology. In conclusion, HIF-P4H-2 inhibition could be a treatment method in retinal neurodegeneration observed in high-pressure glaucoma.